



Review

Ambient air pollution and gestational diabetes mellitus: A review of evidence from biological mechanisms to population epidemiology



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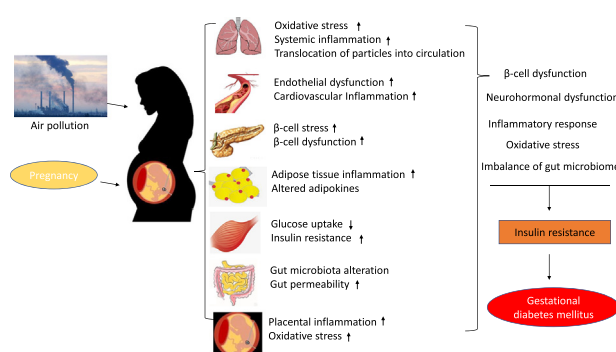
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HIGHLIGHTS

- Potential biological linkage between air pollution exposure and GDM has not been clarified.
- Results of epidemiological studies on ambient air pollution and GDM are inconsistent.
- Meta-analysis showed that first trimester SO₂ increased the risk of GDM while pre-pregnancy O₃ decreased GDM risk.
- No significant effects were observed for PM_{2.5}, PM₁₀ and NO₂.
- Mechanism studies on the molecular level are needed to better understand the relationship.

GRAPHICAL ABSTRACT



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ABSTRACT

Gestational diabetes mellitus (GDM) is a serious complication of pregnancy that could cause adverse health effects on both mothers and fetuses, and its prevalence has been increasing worldwide. Experimental and epidemiological studies suggest that air pollution may be an important risk factor of GDM, but conclusions are inconsistent. To provide a comprehensive overview of ambient air pollution on GDM, we summarized existing evidence concerning biological linkages between maternal exposure to air pollutants and GDM based on mechanism studies. We also performed a quantitative meta-analysis based on human epidemiological studies by searching English databases (Pubmed, Web of Science and Embase) and Chinese databases (Wanfang, CNKI). As a result, the limited mechanism studies indicated that β -cell dysfunction, neurohormonal disturbance, inflammation, oxidative stress, imbalance of gut microbiome and insulin resistance may be involved in air pollution-GDM relationship, but few studies were performed to explore the direct biological linkage. Additionally, a total of 13 epidemiological studies were included in the meta-analysis, and the air pollutants considered included PM_{2.5}, PM₁₀, SO₂, NO₂ and O₃. Most studies were retrospective and mainly conducted in developed regions. The results of meta-analysis indicated that maternal first trimester exposure to SO₂ increased the risk of GDM (standardized odds ratio (OR) = 1.392, 95% confidence intervals (CI): 1.010, 1.773), while pre-pregnancy O₃

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exposure was inversely associated with GDM risk (standardized OR = 0.981, 95% CI: 0.977, 0.985). No significant effects were observed for PM_{2.5}, PM₁₀ and NO₂. In conclusion, additional mechanism studies on the molecular level are needed to provide persuasive rationale underlying the air pollution-GDM relationship. Moreover, other important risk factors of GDM, including maternal lifestyle and road traffic noise exposure that may modify the air pollution-GDM relationship should be considered in future epidemiological studies. More prospective cohort studies are also warranted in developing countries with high levels of air pollution.

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1. Introduction

Gestational diabetes mellitus (GDM) is considered as any degree of hyperglycemia and generally first diagnosed during pregnancy (Mirghani Dirar and Doupis, 2017). According to the International Diabetes Federation (IDF) estimated, GDM affected approximately 14.8% of pregnancies around the world (Federation, 2015). The global prevalence of gestational diabetes mellitus (GDM) is increasing and varied in many areas of the world (Guariguata et al., 2014). From countries with available data, the median prevalence of GDM ranged from 5.8% to 12.9% from 2005 to 2015 (Zhu and Zhang, 2016). It was reported that 7.1%–9.2% of pregnant women with GDM will develop type 2 diabetes mellitus (DM) within 5 years (Lee et al., 2007). The fetus with GDM mother may be at a higher risk to be macrosomia or suffer from cardiometabolic condition in later life (Tam et al., 2017; Yang et al., 2019).

The most studied risk factors of GDM include a family history of diabetes, genetic susceptibility, lifestyle behaviors during pregnancy such as diet, physical activity (Chieffari et al., 2017). With rapid consumption of fossil fuels for transportation, power generation or other human activities, air pollution has become a major public health issue not only in developed countries but also in the developing (Cohen et al., 2017). Current evidence suggests that air pollution may be a risk factor for the development of type 2 DM (Liu et al., 2019a; Rajagopalan and Brook, 2012; Rao et al., 2015). Previous studies have established possible pathological pathways associated

with air pollution, including insulin resistance (Brook et al., 2013; Rao et al., 2015), endothelial dysfunction and systemic Inflammation (Liu et al., 2013; Pope 3rd et al., 2016; Rajagopalan and Brook, 2012). To some extent, GDM shares similar pathogenesis and pathophysiology with type 2 DM (Ben-Haroush et al., 2004), which indicates that ambient air pollution exposure may also affect the development of GDM.

Numerous human epidemiological studies have examined the relationship between air pollution exposure and GDM risk (Choe et al., 2019; Choe et al., 2018; Fleisch et al., 2014; Fleisch et al., 2016; Heejoo et al., 2019; Malmqvist et al., 2013; Padula et al., 2019; Pan et al., 2017; Pedersen et al., 2017; Shen et al., 2017; Zhang et al., 2019). However, conclusions are still inconsistent due to variations among study design, population, exposure evaluation, air pollutant concentrations, as well as GDM diagnosis, which make the results difficult to interpret and provide pregnancy care suggestions to policy makers.

To provide healthcare workers and researchers with more comprehensive estimates of the effect of exposure to ambient air pollution on the risk of GDM, we systematically retrieved all relevant studies to date to summarize the evidence from biological mechanisms to population epidemiology. Understanding the association between air pollution and GDM, elucidating potential mechanisms involved may help to develop preventive measures to promote pregnancy care and decrease the risk of GDM, which is not only beneficial to mothers but also to fetuses.

2. Methods

2.1. Literature search for mechanism studies

We searched mechanism studies from English databases (Pubmed, Web of science and Embase) and three Chinese databases (Wanfang, China National Knowledge Infrastructure, Chongqing VIP Chinese Science and Technology Periodical) with the following keywords: “air pollution”, “outdoor air pollution”, “ambient air pollution”, “fine particulate matter”, “particulate matter 2.5”, “pm2.5”, “pm10”, “nitrogen dioxide”, “NO₂”, “NO_x”, “ozone”, “SO₂”, “sulfur dioxide”, “carbon monoxide”, “diabetes mellitus”, “diabetes”, “gestational diabetes”, “pregnancy diabetes”, “gestational diabetes mellitus”, “ β -cell dysfunction”, “neurohormonal disturbance”, “inflammation”, “oxidative stress”, “imbalance of gut microbiome”, “insulin resistance”. We limited our search to articles published up to 24 September 2019. We reviewed mechanism studies attempted to investigate the pathophysiological response for air pollution and diabetes. Then we summarized the potential biological linkage between air pollution induced pathophysiological reaction and the pathophysiology of GDM to illustrate the mechanisms underlying air pollution and the development of GDM.

2.2. Literature search for epidemiological studies

We followed the standard systematic review methods and reported the required items according to the guideline of the Preferred Reporting Items for Systematic Review and Meta-analyses criteria (Bernardo, 2017). Publications were searched from three English databases

(Pubmed, Web of science and Embase) and three Chinese databases (Wanfang, China National Knowledge Infrastructure, Chongqing VIP Chinese Science and Technology Periodical) up to September 24, 2019. The search strategy and results are summarized in Supplementary Table 1. Data selection was conducted by two independent researchers according to the inclusion and exclusion criteria.

2.2.1. Selection criteria for systematical review and meta-analysis

Publications from different databases were extracted after screening titles and abstracts in the scope of this study. We included studies if they met the following criteria: (1) Studies in the field of exposing to ambient air pollution and GDM; (2) GDM diagnosed with standardized criteria by physicians or had clear medical records; (3) Reported quantitative effect estimates and their 95% confidence intervals (CIs). We also excluded studies if they are: (1) Review documents; (2) Animal studies; (3) Repeat literature; (4) Studies concerning of type 1 or 2 DM; (5) Without clear increment of air pollutant. Finally, considering the above criteria, this meta-analysis was synthesized based on 13 studies from six databases. The search procedure is shown in Fig. 1.

2.2.2. Study quality assessment

We evaluated the quality of included studies using the Newcastle-Ottawa Scale (NOS) (Li et al., 2017; Wells et al., 2013). The results are shown in Supplementary Table 2. The following items were evaluated: (1) Representativeness of the longitudinal study; (2) Selection of the non-exposed cohort; (3) Ascertainment of air pollution exposure; (4) Demonstration that if outcome of interest is present at start of study or not; (5) Comparability of cohorts on the basis of the design

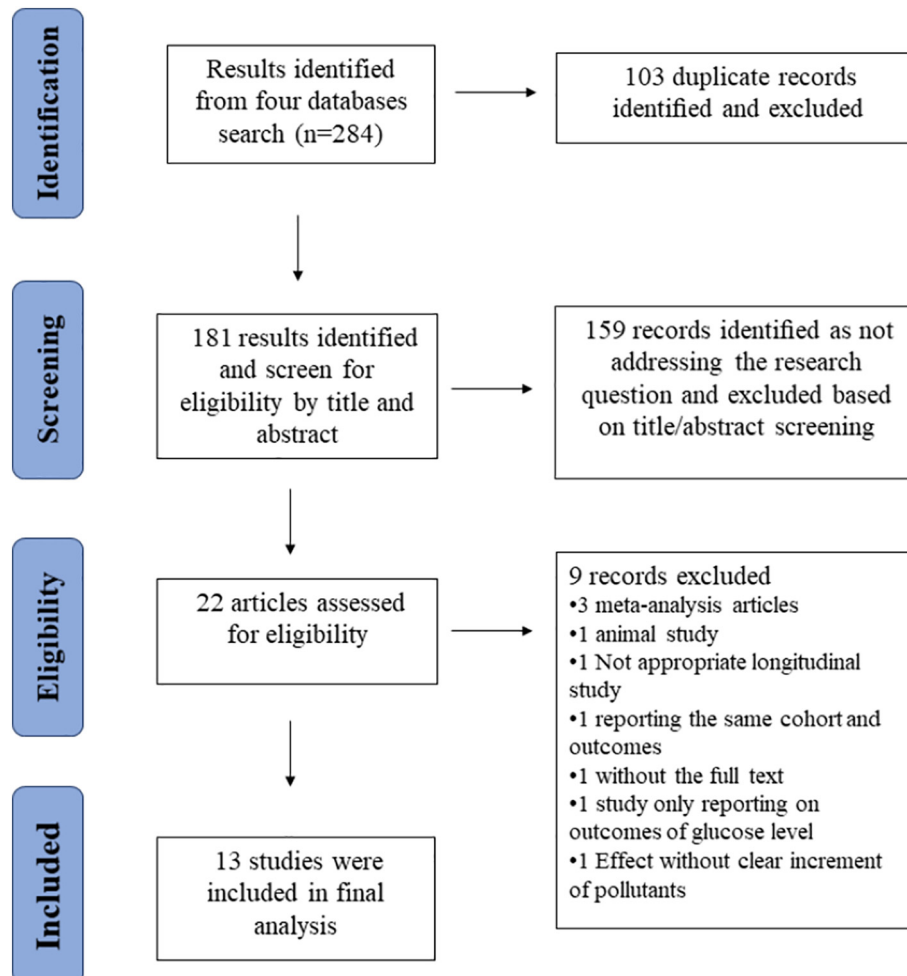


Fig. 1. Process of literature search.

or analysis; (6) Reliability of GDM diagnosis; (7) Follow-up long enough for outcome to occur; (8) Adequacy of follow up of cohort.

2.2.3. Data extraction

Information was extracted including the below items by two researchers: Author (Publication year), location, study period, data source, sample size, GDM cases, diagnosis of GDM, pollutants, assignment of exposure, exposure period and the effect estimates (ORs and 95% CIs). Details are shown in Tables 1 and 2.

2.2.4. Statistical analysis

This meta-analysis was conducted to examine association between air pollution exposure and GDM risk quantitatively. All estimated effects of selected studies included odds ratios (OR). One study (Robledo et al., 2015) estimated relative risk (RR), it was roughly considered equivalent to OR. Considering most studies examined the association using single pollutant models, the effect estimates from single-pollutant models or the “primary analyses” were extracted. The amount of studies was limited (<2) for some pollutants, including black carbon, CO, NO or NO_x, so that the pooled estimates were only exhibited for PM_{2.5}, PM₁₀, SO₂, NO₂ and O₃. Prior to performing the meta-analysis, studies expressing SO₂,

NO₂ and O₃ concentrations as “ppb” were converted to the unit of µg/m³ (Liu et al., 2019a): 1 ppb = M/22.4 µg/m³, where M is the molecular weight of SO₂, NO₂ or O₃. Standardized increments (10 µg/m³ in all pollutants) were used by calculating the following formula (Liu et al., 2019a):

$$OR_{(Standardized)} = OR_{(Original)}^{\text{Increment}(10)/\text{Increment}(Original)}$$

Most studies used an interquartile range (IQR) as the increment, we chose the effects estimated using an IQR in the primary meta-analysis. Fleisch et al. assessed air pollution exposure based on a fixed central monitoring station and the spatiotemporal model (Fleisch et al., 2014), respectively. To reduce potential exposure misclassification, we extracted the estimated effects based on spatiotemporal model from this study. One study from Denmark used both Danish clinical guidelines and WHO criteria to identify GDM (Pedersen et al., 2017). Due to the WHO criterion have been rarely used in Denmark, we extracted effect estimates from the Danish clinical guidelines in this study. The random-effect model would be conducted if $P < 0.1$ and $I^2 > 50\%$, otherwise, we selected a fix-effect model. All the estimated effects were

Table 1
Epidemiological studies on the association between air pollution exposure and GDM.

Author (publication year)	Location	Study period	Data source	Sample size	GDM cases	Diagnosis of GDM	Pollutants	Assignment of exposure	Exposure period
Malmqvist et al. (2013)	Sweden	1999–2005	Swedish Medical Birth Registry	81,110	1599	WHO criteria	NO _x	Gaussian flat two-dimensional dispersion model	Entire pregnancy
Fleisch et al. (2014)	US	1999–2002	Women recruited in Boston-area at their first prenatal visit to Harvard Vanguard Medical Associates	2093	118	American Diabetes Association	PM _{2.5} , BC	Spatiotemporal land use regression model	Second trimester
Hu et al. (2015)	US	2004–2005	Bureau of Vital Statistics and Office of Health Statistics and Assessment, Florida Department of Health	410,267	14,032	American Diabetes Association	PM _{2.5} , O ₃	Hierarchical Bayesian space-time statistical model	First trimester, second trimester, the full pregnancy
Robledo et al. (2015)	US	2002–2008	A retrospective cohort study from the Consortium on Safe Labor	219,952	11,334	American Diabetes Association	PM _{2.5} , PM ₁₀ , NO _x , CO, SO ₂ , O ₃	CMAQ	3 months prior to conception, first trimester
Fleisch et al. (2016)	US	2003–2008	Massachusetts Registry of Vital Records and Statistics	159,373	5381	American Diabetes Association	PM _{2.5}	Hybrid satellite-based spatiotemporal model	First trimester, second trimester
Pan et al. (2017)	Taiwan, China	2004–2005	Taiwan's Birth Registration Database	19,606	378	American Diabetes Association	PM ₁₀ , CO, NO, NO ₂ , NO _x , SO ₂ , O ₃	Spatial interpolation method (i.e., ordinary kriging)	First, second and third trimester
Pedersen et al. (2017)	Denmark	1997–2002	Danish National Birth Cohort	72,745	565	Danish clinical guidelines and WHO criteria	NO ₂	Advanced AirGIS dispersion model	First, second, third trimester and full pregnancy
Shen et al. (2017)	Taiwan, China	2006–2013	Taiwan's National Health Insurance Research Data	1,000,000	11,688	International Classification of Diseases criteria	PM _{2.5} , SO ₂ , O ₃ , NO ₂	Spatial interpolation method (i.e., ordinary kriging)	12-week prior to pregnancy, first trimester, second trimester
Choe et al. (2018)	US	2002–2012	National Perinatal Information Center	61,640	4884	International Classification of Diseases criteria	PM _{2.5} and BC	Both land use regression and satellite remote sensing method	First and second trimester
Choe et al. (2019)	US	2008–2010	New York State Department of Health Statewide Planning and Research Cooperative System	256,372	17,065	International Classification of Diseases criteria	PM _{2.5} , NO ₂	Spatiotemporal model	First and second trimester
Heejoo et al. (2019)	US	1999–2009	Kaiser Permanente Southern California (KPSC) hospitals	239,574	18,244	American Diabetes Association	PM _{2.5} , PM ₁₀ , NO ₂ , O ₃	Inverse distance-weighted method	Preconception and first trimester
Yao et al. (2019)	Mainland, China	2015–2018	Prospective cohort study from Hefei, China	4817	1030	Chinese GDM criteria	PM _{2.5} , PM ₁₀ , SO ₂ , NO ₂ , CO	Fixed monitoring stations	Preconception and first trimester
Zhang et al. (2019)	Mainland, China	2011–2014	Prospective cohort study from Guangzhou, China	5165	604	IADPSG criteria	PM _{2.5} , PM ₁₀ , SO ₂ , NO ₂	Inverse distance-weighted method	First and second trimester

BC: black carbon; CMAQ: community multi-scale air quality model; WHO: World Health Organization; IADPSG: International Association of Diabetes and Pregnancy Study Groups.

Table 2

Summary of OR estimates (95%CI) for included studies of the association between air pollution exposure and risk of GDM.

Author (publication year)	Effect estimates (95% CI)		
Malmqvist et al. (2013)	NO _x (μg/m ³) Q1 = 2.5–8.9 μg/m ³ Q2 = 9.0–14.1 μg/m ³ Q3 = 14.2–22.6 μg/m ³ Q4 ≥ 22.7 μg/m ³	First trimester 1.00 (Reference) 1.27 (1.05–1.53) 1.48 (1.23–1.77) 1.65 (1.38–1.99)	Second trimester 1.00 (Reference) 1.19 (0.99–1.44) 1.52 (1.28–1.82) 1.69 (1.41–2.03)
Fleisch et al. (2014)	Per IQR (2.0 μg/m ³) increase for PM _{2.5} : 0.94 (0.67–1.34) in second trimester Per IQR (0.34 μg/m ³) increase for BC: 1.02 (0.73–1.41) in second trimester		
Hu et al. (2015)	First trimester Per 5 μg/m ³ increase for PM _{2.5} : 1.16 (1.11–1.21) Per 5 ppb increase for O ₃ : 1.09 (1.07–1.11)	Second trimester Per 5 μg/m ³ increase for PM _{2.5} : 1.15 (1.10–1.20) Per 5 ppb increase for O ₃ : 1.12 (1.10–1.14)	Full pregnancy Per 5 μg/m ³ increase for PM _{2.5} : 1.20 (1.13–1.26) Per 5 ppb increase for O ₃ : 1.18 (1.15–1.21)
Robledo et al. (2015)	Preconception Per IQR (5.54 μg/m ³) increase for PM _{2.5} : 0.97 (0.93–1.02) Per IQR (3.30 ppb) increase for SO ₂ : 1.05 (1.01–1.09) Per IQR (12.33 ppb) increase for O ₃ : 0.93 (0.90–0.96) Per IQR (6.30 μg/m ³) increase for PM ₁₀ : 0.99 (0.96–1.02)	First trimester Per IQR (5.28 μg/m ³) increase for PM _{2.5} : 0.98 (0.94–1.04) Per IQR (3.31 ppb) increase for SO ₂ : 1.04 (1.00–1.08) Per IQR (12.36 ppb) increase for Sulfate: 1.00 (0.97–1.03) Per IQR (6.32 μg/m ³) increase for PM ₁₀ : 0.98 (0.95–1.01)	
Fleisch et al. (2016)	First trimester Per 4.3 μg/m ³ increase for PM _{2.5} : 1.01 (0.93–1.09)	Second trimester Per 4.5 μg/m ³ increase for PM _{2.5} : 0.97 (0.90–1.05)	
Pan et al. (2017)	First trimester Per 10 μg/m ³ increase for PM ₁₀ : 0.99 (0.93–1.05) 0.1 ppm increment for CO: 1.08 (1.00–1.15) Per ppb for NO: 1.05 (1.02–1.08) Per ppb for NO ₂ : 1.01 (0.99–1.03) Per ppb for NO _x : 1.02 (1.00–1.03) Per ppb for SO ₂ : 1.05 (1.00–1.11) Per ppb for O ₃ : 0.97 (0.94–0.99)	Second trimester Per 10 μg/m ³ increase for PM ₁₀ : 0.96 (0.90–1.03) 0.1 ppm increment for CO: 1.06 (0.99–1.14) Per ppb for NO: 1.05 (1.02–1.08) Per ppb for NO ₂ : 1.01 (0.99–1.03) Per ppb for NO _x : 1.01 (1.00–1.03) Per ppb for SO ₂ : 1.06 (1.00–1.12) Per ppb for O ₃ : 0.97 (0.94–1.00)	Third trimester Per 10 μg/m ³ increase for PM ₁₀ : 0.95 (0.90–1.00) 0.1 ppm increment for CO: 1.02 (0.95–1.09) Per ppb for NO: 1.03 (1.00–1.06) Per ppb for NO ₂ : 1.00 (0.98–1.02) Per ppb for NO _x : 1.01 (0.99–1.02) Per ppb for SO ₂ : 1.02 (0.97–1.09) Per ppb for O ₃ : 0.98 (0.96–1.01)
Pedersen et al. (2017)	First trimester	Second trimester	Third trimester
Danish clinical guidelines	Per 10 μg/m ³ for NO ₂ : 0.89 (0.76–1.03)	Per 10 μg/m ³ for NO ₂ : 0.91 (0.78–1.06)	Per 10 μg/m ³ for NO ₂ : 0.95 (0.82–1.10)
WHO criteria	Per 10 μg/m ³ for NO ₂ : 1.24 (1.03–1.49)	Per 10 μg/m ³ for NO ₂ : 1.25 (1.05–1.51)	Per 10 μg/m ³ for NO ₂ : 1.29 (1.08–1.55)
Shen et al. (2017)	Within the 12-week prior to pregnancy Per IQR (13.25 μg/m ³) increase for PM _{2.5} : 1.10 (1.03–1.18) Per IQR (1.62 ppb) increase for SO ₂ : 1.37 (1.30–1.45) Per IQR (6.22 ppb) increase for O ₃ : 1.01 (0.96–1.07) Per IQR (7.51 ppb) increase for NO ₂ : 0.96 (0.90–1.02)	First trimester Per IQR (12.86 μg/m ³) increase for PM _{2.5} : 1.09 (1.02–1.17) Per IQR (1.60 ppb) increase for SO ₂ : 1.37 (1.30–1.45) Per IQR (6.28 ppb) increase for O ₃ : 1.02 (0.96–1.08) Per IQR (7.52 ppb) increase for NO ₂ : 0.93 (0.88–1.00)	Second trimester Per IQR (12.86 μg/m ³) increase for PM _{2.5} : 1.07 (1.01–1.14) Per IQR (1.54 ppb) increase for SO ₂ : 1.38 (1.31–1.46) Per IQR (6.26 ppb) increase for O ₃ : 1.04 (0.99–1.11) Per IQR (7.50 ppb) increase for NO ₂ : 0.97 (0.93–1.02)
Choe et al. (2018)	First trimester Per IQR (4.31 μg/m ³) increase for PM _{2.5} : 1.02 (0.95, 1.09) Per IQR (0.22 μg/m ³) increase for BC: 1.03 (0.98, 1.08)	Second trimester Per IQR (4.08 μg/m ³) increase for PM _{2.5} : 1.08 (1.00, 1.15) Per IQR (0.23 μg/m ³) increase for BC: 1.01 (0.97, 1.06)	
Choe et al. (2019)	First trimester Per IQR (3.23 μg/m ³) increase for PM _{2.5} : 0.97 (0.94–1.00) Per IQR (7.96 ppb) increase for NO ₂ : 1.05 (1.01–1.09)	Second trimester Per IQR (3.23 μg/m ³) increase for PM _{2.5} : 1.06 (1.03–1.10) Per IQR (7.96 ppb) increase for NO ₂ : 1.02 (0.98–1.06)	
Heejoo et al. (2019)	Preconception Per IQR (6.5 μg/m ³) increase for PM _{2.5} : 1.04 (1.01–1.06) Per IQR (16.1 μg/m ³) increase for PM ₁₀ : 1.03 (1.00–1.06) Per IQR (10.4 ppb) increase for NO ₂ : 1.10 (1.07–1.13) Per IQR (15.7 ppb) increase for O ₃ : 0.94 (0.92–0.95)	First trimester Per IQR (6.5 μg/m ³) increase for PM _{2.5} : 0.98 (0.95–1.00) Per IQR (16.1 μg/m ³) increase for PM ₁₀ : 0.98 (0.95–1.01) Per IQR (10.4 ppb) increase for NO ₂ : 1.02 (0.99–1.05) Per IQR (15.7 ppb) increase for O ₃ : 0.95 (0.94–0.97)	
Yao et al. (2019)	Preconception Per 10 μg/m ³ increase for PM _{2.5} : 1.14 (1.08–1.20) Per 10 μg/m ³ increase for PM ₁₀ : 1.13 (1.08–1.19) Per 1 μg/m ³ increase for SO ₂ : 1.03 (1.01–1.05) Per 0.1 mg/m ³ increase for CO: 1.07 (1.01–1.13)	First trimester Per 10 μg/m ³ increase for PM _{2.5} : 0.99 (0.92–1.05) Per 10 μg/m ³ increase for PM ₁₀ : 1.02 (0.96–1.09) Per 1 μg/m ³ increase for SO ₂ : 1.02 (1.01–1.05) Per 0.1 mg/m ³ increase for CO: 1.01 (0.94–1.08)	
Zhang et al. (2019)	First trimester Per 21.41 μg/m ³ increase for PM _{2.5} : 1.13 (0.84–1.52) Per 21.74 μg/m ³ increase for PM ₁₀ : 1.09	Second trimester Per 20.64 μg/m ³ increase for PM _{2.5} : 0.83 (0.47–1.49) Per 26.46 μg/m ³ increase for PM ₁₀ : 0.81	

(continued on next page)

Table 2 (continued)

Author (publication year)	Effect estimates (95% CIs)
	(0.74–1.61)
	Per 3.08 $\mu\text{g}/\text{m}^3$ increase for SO_2 : 1.22 (1.02–1.47)
	Per 16.21 $\mu\text{g}/\text{m}^3$ increase for NO_2 : 1.17 (0.93–1.48)
	(0.46–1.44)
	Per 2.75 $\mu\text{g}/\text{m}^3$ increase for SO_2 : 0.93 (0.78–1.11)
	Per 20.23 $\mu\text{g}/\text{m}^3$ increase for NO_2 : 0.65 (0.41–1.02)

BC: Black carbon; IQR: interquartile range; SD: standard deviation. ppb: parts per billion.

grouped by different exposure period (pre-pregnancy, first trimester, second trimester or third trimester). In the primary meta-analysis, we included estimated effects based on the increment of IQR from Shen et al.'s study (Shen et al., 2017), we also meta-analyzed estimated effects based on the increment of SD extracted from this study as the sensitivity analysis. Begg's test as well as Egger's test were conveyed to examine publication bias (Stieb et al., 2012), we considered a publication bias if $P < 0.05$. All these analyses were performed with Stata (Version 14.0).

3. Results

3.1. Evidence on the biological mechanisms of air pollution-GDM relationship

The involved pathophysiology of air pollution-mediated GDM remain unclear. Nevertheless, the following plausible mechanisms are believed to be involved in air pollution and the development of GDM (Fig. 2).

3.1.1. β -Cell dysfunction

β -Cells in pancreas has functions of storing and secreting insulin to response to glucose load. β -cell dysfunction is known as β -cells cannot sense glucose concentration in the circulation effectively or generate enough insulin to maintain glucose homeostasis (Plows et al., 2018). Deficiencies in β -cell machinery may appear during pregnancy (Pasek et al., 2016). β -Cell failure could impact insulin secretion, which contributes to the development of hyperglycemia (Ashcroft et al., 2017; Delghingaro-Augusto et al., 2009). Further, glucotoxicity might result in β -cell apoptosis (Ashcroft et al., 2017). The reduced β -cell mass and β -cell number combined with β -cell dysfunction might promote the process of GDM (Plows et al., 2018). It was suggested that GDM was mostly related with an exacerbation of β cell dysfunction (Agha-Jaffar et al., 2016; Batchuluun et al., 2018; Li et al., 2014). Alderete et al. suggested that exposure to NO_2 and $\text{PM}_{2.5}$ negatively affected pancreatic β -cell function among children and eventually lead to type 2 DM (Alderete et al., 2017). Deficiency in the β -cell machinery exists in pregnancy (Chiefari et al., 2017), further exposure to ambient air pollution might impair β -cell function, thus air pollution may lead to GDM by accelerating β -cell dysfunction among pregnant women.

3.1.2. Neurohormonal dysfunction

Imbalance of adipokines, important regulators of neurohormonal metabolic control, may aggravate insulin resistance and promote GDM onset (Bellos et al., 2019; de Gennaro et al., 2019; Kajimura, 2017). Leptin (LEP) and adiponectin are the most important regulators of neurohormonal metabolic control (Plows et al., 2018). Xu et al. observed that long-term exposure to $\text{PM}_{2.5}$ decreased adiponectin and leptin in C57BL/6 mice after long term exposure to $\text{PM}_{2.5}$ (6 h/day, 5 days/week for duration of 10 months) (Xu et al., 2011). Lavigne et al. found that an IQR increase in average exposure to $\text{PM}_{2.5}$ and NO_2 during pregnancy was associated with an 11% (95% CI: 4–17) and 13% (95% CI: 6, 20) increase in adiponectin levels, respectively (Lavigne et al., 2016). Alderete et al. found one standard deviation (1-SD: 2 ppb) increase in prenatal non-freeway NO_x was associated with 33% ($P = 0.01$) higher

leptin and 9% higher adiponectin levels ($P = 0.07$) in cord blood (Alderete et al., 2018). We posit that air pollution might cause inflammatory responses in white adipose tissues, resulting in an imbalance of adipokines or leptins, which subsequently exacerbate the development of GDM.

3.1.3. Inflammation

Studies in vivo and vitro indicated that exposure to air pollution can result in local or systemic inflammation through multiple pathways (Chen et al., 2018; Guan et al., 2019; Guan et al., 2018; Lee et al., 2018; Ogino et al., 2017; Ostro et al., 2014; Pope 3rd et al., 2016; Wang et al., 2018a), which produce proinflammatory mediators, including $\text{TNF-}\alpha$, IL-1 β , IL-6 and IL-8 (Hogg and van Eeden, 2009). Sun et al. firstly observed systemic inflammation in C57BL/6 mice when exposure to ambient $\text{PM}_{2.5}$ for 24 weeks (6 h/day, 5 days/week), the plasma concentration of $\text{TNF-}\alpha$, IL-6, E-selectin, intracellular adhesion molecule-1 (ICAM-1), plasminogen activator inhibitor-1 were significantly increased (Sun et al., 2009). Liu et al. observed that $\text{PM}_{2.5}$ exposure can regulate visceral adipose tissue inflammation by impairing adenosine monophosphate activated protein kinase (AMPK) and Akt signaling via both CC-chemokine receptor 2-dependent and -independent pathways in C57BL/6 mice (Liu et al., 2014). Human epidemiological studies have observed that cellular and inflammatory mediators, such as IL-1 β , IL-6, IL-8, IFN- γ , C-reactive protein, $\text{TNF-}\alpha$, fibrinogen, and white blood cells increased after acute and long-term exposure to air pollution (Chen et al., 2018; Delfino et al., 2010; Guan et al., 2018; Wu et al., 2018; Yang et al., 2017; Zhang et al., 2018). The elevated inflammatory response plays an important role in development of pregnancy complications, including GDM (Abell et al., 2015; Kuzmicki et al., 2009; Morisset et al., 2011). Air pollution may trigger the release of inflammatory cytokines when pregnant women exposure to air pollution, then the elevated inflammatory response subsequently contributes to insulin resistance, which consequently promotes the development of GDM.

3.1.4. Oxidative stress

In general, pregnant women are physiologically hyperglycemia, which might induce oxidative stress through several metabolic mechanisms, such as enhanced reactive oxygen species (ROS) production in the mitochondria, polyol pathway, formation of advanced glycation end-products (AGE) and activation of protein kinase C (PKC) (Lappas et al., 2011). Oxidative stress can decrease glucose transporter type 4 (GLUT4) expression by impairing nuclear proteins to the insulin responsive element in the GLUT4 promoter (Pessler et al., 2001). Oxidative stress activates NF- κB translocation into the nucleus, which can cause inflammatory response (Lappas et al., 2009). Particulate matters are enriched in metals, sulfur and organic components (Jiang et al., 2019; Reddy et al., 2005), which was thought to induce the oxidative damage (Lui et al., 2019). In C57BL/6 mice model, Haberkett et al. found short-term exposure to $\text{PM}_{2.5}$ increased vascular insulin resistance and inflammation induced by pulmonary oxidative stress (Haberkett et al., 2016). In vitro, Gram-negative *Escherichia coli* (*E. coli*) coliform bacterium were exposed to extracted PM samples, the intracellular ROS production increased (Velali et al., 2019). Epidemiology evidence showed that chronic air

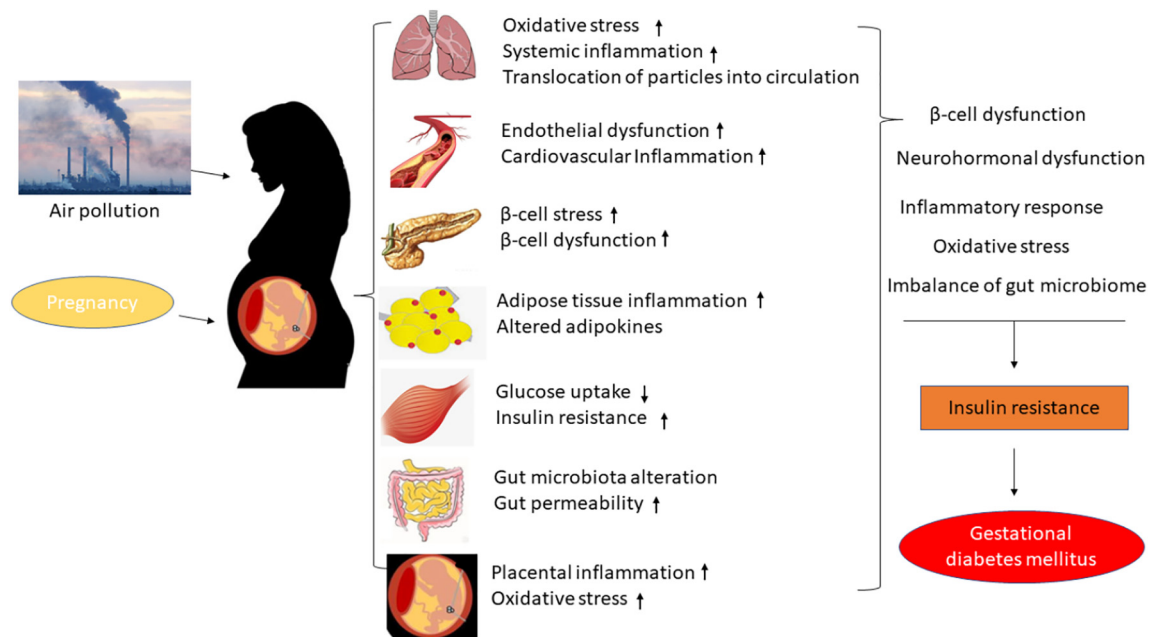


Fig. 2. Mechanistic insight into air pollution exposure and the development of GDM.

pollution exposure can increase oxidative stress among pregnant women, the levels of malondialdehyde, glutathione and superoxide dismutase increased (Nagiah et al., 2015). Both experimental and epidemiological studies are plausible in indicating that oxidative stress is an important causal pathway in the associations between air pollution exposure and increased risk of GDM.

3.1.5. Gut microbiome

Emerging evidence suggests that gut microbiome is associated with metabolic diseases, including GDM (Plows et al., 2018). Fugmann et al. reported a lower proportion of the phylum Firmicutes and a higher proportion of the family *Prevotellaceae* among women with previous GDM (Fugmann et al., 2015). *Prevotellaceae* mainly contributed to increasing gut permeability and was regulated by tight junction proteins, including zonulin-1 (ZO-1). Mekkala et al. found that increased “free” plasma/serum ZO-1 was associated with GDM (Mekkala et al., 2017). What's more, the elevation of gut permeability may promote movement of inflammatory mediators from gut into circulation, which subsequently elevates the level of inflammatory cytokines in the body (Chow et al., 2010; Jayashree et al., 2014). Recently, Mutlu et al. found that exposure to particulate matter altered the gut microbiota in C57BL/6 mice model, a potential mechanism to explain the inflammation effect induced by PM (Mutlu et al., 2018). Liu et al. demonstrated that the associations of PM_{2.5} and PM₁ with the risk of type 2 DM were partially mediated by the gut microbiota diversity in adults (Liu et al., 2019b). Air pollution may also adversely impact gut permeability by altering the microbiota among pregnant women. Therefore, the possible mechanism may be that gut permeability increased when pregnant women exposure to air pollution during pregnancy, which subsequently accelerates the active transport of inflammatory cytokines from the gut into the circulation. The elevated inflammatory state then accelerates the development of insulin resistance, which subsequently contributes to the risk of GDM.

3.1.6. Insulin resistance

Insulin resistance is known as cells cannot respond to insulin effectively. Current epidemiological evidence suggests that insulin resistance could be affected by air pollution (Brook et al., 2013; Kim and Hong, 2012; Thiering et al., 2013; Wolf et al., 2016). In C57BL/6 mice model, PM_{2.5} exposure (6 h/day, 5 days/week for duration of 10 months) induced insulin resistance in adipose tissues, liver, and skeletal muscle

(Xu et al., 2011). It was reported that insulin resistance induced by air pollution was one of the important underlying metabolic conditions predisposing to the development of type 2 DM (Rajagopalan and Brook, 2012; Rao et al., 2015). Long term exposure to air pollution could lead to insulin resistance, then provoke metabolic dysfunction (Rajagopalan and Brook, 2012). In the Korean Elderly Environmental Panel (KEEP) Study, Kim et al. found that air pollutants such as PM₁₀ and NO₂ were positively associated with insulin serum level and homeostasis model assessment (HOMA) index, which was generally used to test the level of insulin resistance (Kim and Hong, 2012). In Iran, Kelishadi et al. conducted a study of 374 children in age of 10–18 years old and found insulin resistance was significantly associated with Pollution Standard Index (PSI) (Kelishadi et al., 2009). In another prospective German birth cohort study conducted among 10-year-old children, for every 2 standard deviations increase in ambient NO₂ and PM₁₀, level of insulin resistance increased by 17.0% (95% CI 5.0–30.3) and 18.7% (95% CI 2.9–36.9), respectively (Thiering et al., 2013). Insulin resistance is a critical pathogenesis of GDM (Plows et al., 2018). Exposure to air pollution may lead to the development of insulin resistance among pregnant women, then accelerate the development of GDM.

3.2. Evidence on the epidemiological association between air pollution and GDM

3.2.1. Study characteristics

A total of 284 records were identified in the original search. After removing duplicates, we identified 181 articles for reviewing titles and abstracts (Fig. 2). Finally, 13 full-text articles met the inclusion criteria and were included in our meta-analysis (Choe et al., 2019; Choe et al., 2018; Fleisch et al., 2014; Fleisch et al., 2016; Heejoo et al., 2019; Hu et al., 2015; Malmqvist et al., 2013; Pan et al., 2017; Pedersen et al., 2017; Robledo et al., 2015; Shen et al., 2017; Yao et al., 2019; Zhang et al., 2019). Of these 13 articles, 7 studies reported results from United States (Choe et al., 2019; Choe et al., 2018; Fleisch et al., 2014; Fleisch et al., 2016; Heejoo et al., 2019; Hu et al., 2015; Robledo et al., 2015), two from Taiwan (Pan et al., 2017; Shen et al., 2017), one from Sweden (Malmqvist et al., 2013), one study from Denmark (Pedersen et al., 2017) and two from China mainland (Yao et al., 2019; Zhang et al., 2019). The sample size ranged from 2093 to 1 million. Two studies investigated the exposure of entire pregnancy, which was from date of

conception to the delivery of each woman (Hu et al., 2015; Pedersen et al., 2017). Two studies analyzed the effect of air pollution exposure in the third trimester (Pan et al., 2017; Pedersen et al., 2017). The study period varied from 2 to 11 years. A wide range of air pollutants, including PM_{2.5}, PM₁₀, SO₂, NO₂ and O₃, were examined in current studies. Different increments of pollutants were used to calculate the effect estimates in these selected studies, including the IQR or other fixed increments (Table 2). More details are shown in Tables 1 and 2. All studies concerning air pollution and GDM were of high quality (score ranged from 7 to 8 in NOS) (Supplementary Table 2).

3.2.2. Exposure assessment

A variety of models were used to assess air pollution exposure, including Community Multi-scale Air Quality model (CMAQ) (Robledo et al., 2015), hierarchical Bayesian space-time statistical model (Hu et al., 2015), spatial interpolation method (Pan et al., 2017; Shen et al., 2017), spatiotemporal models (Choe et al., 2019; Choe et al., 2018; Fleisch et al., 2014; Fleisch et al., 2016), inverse distance-weighted model (Heejoo et al., 2019; Zhang et al., 2019), advanced AirGIS dispersion model (Pedersen et al., 2017), Gaussian flat two-dimensional dispersion model (Malmqvist et al., 2013). Yao et al. did not provide the details of air pollution exposure assessment model (Yao et al., 2019). Most studies reported the averaged air pollutant concentration in different exposure period, including pre-pregnancy period, the first, second and third trimester. The ranges of each trimester differed and were defined as follows (1): Pre-pregnancy period (including 91 days (Robledo et al., 2015) or 12-week prior to pregnancy (Heejoo et al., 2019; Shen et al., 2017; Yao et al., 2019)); (2): First trimester (from 1st to 12th

weeks of gestation (Choe et al., 2019; Choe et al., 2018; Fleisch et al., 2016; Heejoo et al., 2019; Pan et al., 2017; Shen et al., 2017; Yao et al., 2019; Zhang et al., 2019) or from 1st to 13th weeks of gestation (Hu et al., 2015; Robledo et al., 2015)); (3): Second trimester (from 13th to 24th weeks of gestation (Fleisch et al., 2016; Shen et al., 2017), 13th to 26th weeks of gestation (Choe et al., 2019; Choe et al., 2018; Pan et al., 2017), 13th to 27th weeks of gestation (Zhang et al., 2019), or from 14th to 26th weeks of gestation (Hu et al., 2015)); (4): Third trimester (from 27th week of gestation to birth (Pan et al., 2017)). All effect estimates for each selected pollutant were pooled by different exposure period in the meta-analysis (pre-pregnancy, first, second or third trimester).

3.2.3. Pooled estimates of ambient air pollution exposure on GDM

The pooled effects of PM_{2.5}, PM₁₀, SO₂, NO₂ and O₃ on the risk of GDM were meta-analyzed in different exposure period (Figs. 3–7). A 10 µg/m³ increase in first trimester SO₂ (standardized OR = 1.392, 95% CI: 1.010, 1.773) was associated with GDM. Pre-pregnancy (standardized OR = 1.450, 95% CI: 0.878, 2.023) or second trimester (standardized OR = 1.383, 95% CI: 0.666, 2.101) SO₂ was not associated with GDM risk (Fig. 5). Heterogeneity were significant among studies in different trimesters (I^2 = 96.7%, 93.6% and 93.6%, respectively) (Fig. 5).

Fig. 7 presents the pooled estimates for GDM associated with an increment of 10 µg/m³ in O₃ exposure. We observed that pre-pregnancy exposure to O₃ was inversely associated with GDM risk (standardized OR = 0.981, 95% CI: 0.977, 0.985) (Fig. 7). No significant association was found in first (standardized OR = 1.007, 95% CI: 0.970, 1.045) or

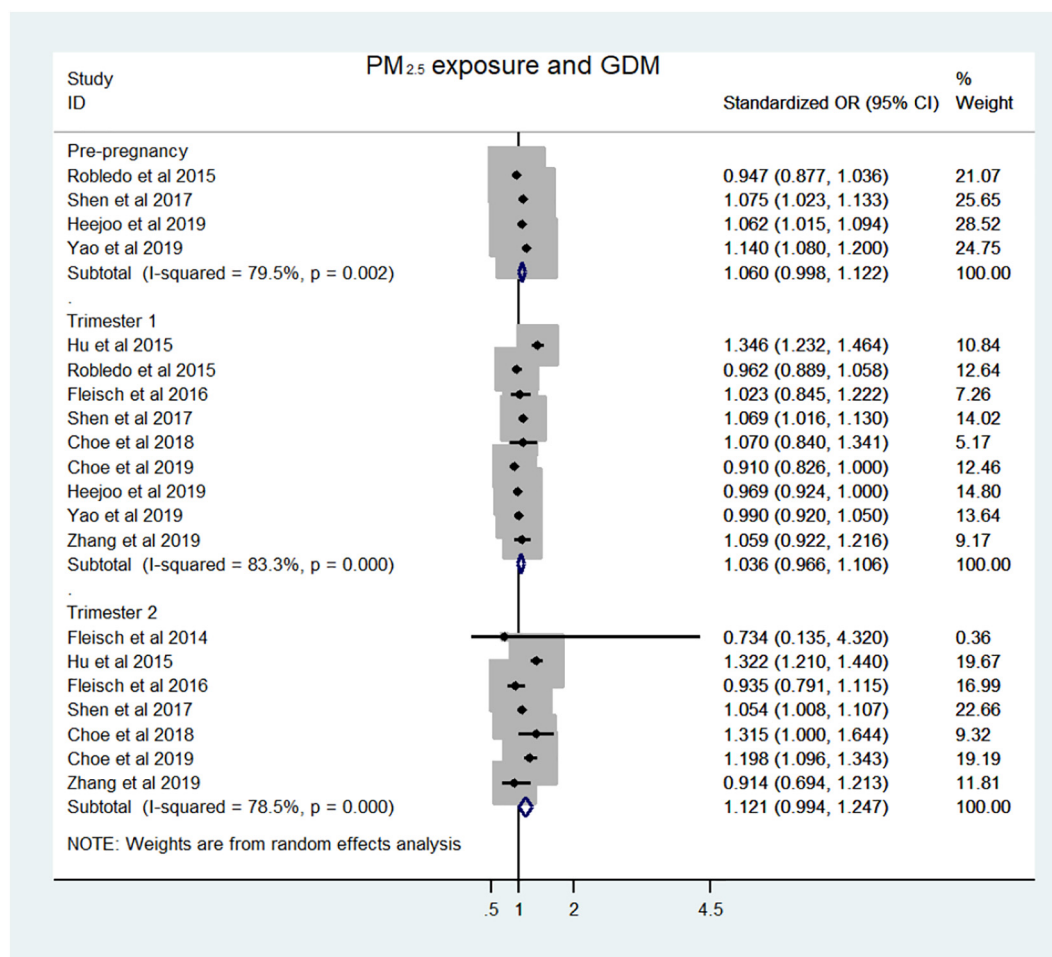


Fig. 3. Standardized odds ratio (OR) and 95% confidence intervals (CI) of GDM associated with 10 µg/m³ increase in PM_{2.5} by gestational period; pooled estimates of effect size are indicated by vertical points of diamonds and 95% CI are represented by horizontal points; size of shaded area around point estimate is proportional to weight in calculating pooled estimate.

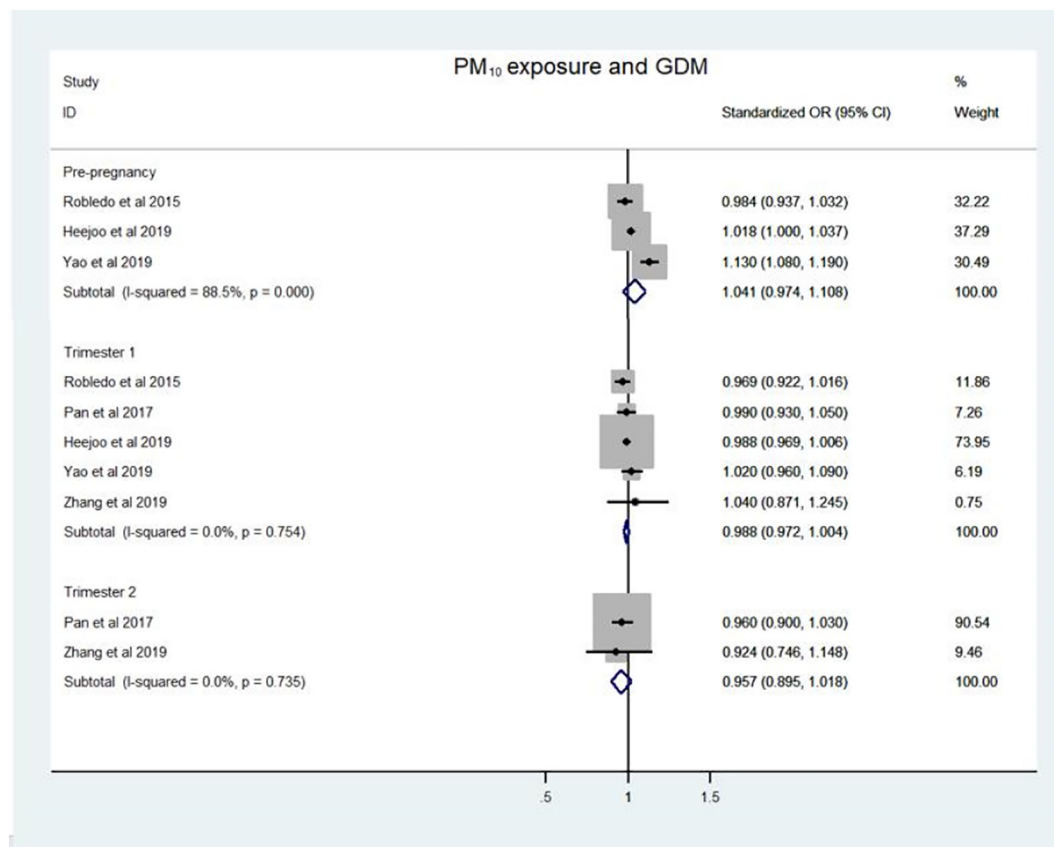


Fig. 4. Standardized odds ratio (OR) and 95% confidence intervals (CI) of GDM associated with $10 \mu\text{g}/\text{m}^3$ increase in PM_{10} by gestational period; pooled estimates of effect size are indicated by vertical points of diamonds and 95% CI are represented by horizontal points; size of shaded area around point estimate is proportional to weight in calculating pooled estimate.

second trimester (standardized OR = 1.024, 95% CI: 0.929, 1.119). High heterogeneity among studies in first and second trimester were observed ($I^2 = 96.4\%$ and 91.7%).

No significant association was observed in pre-pregnancy (standardized OR = 1.060, 95% CI: 0.998, 1.122), first trimester (standardized OR = 1.036, 95% CI: 0.966, 1.106) or second trimester $\text{PM}_{2.5}$ (standardized OR = 1.121, 95% CI: 0.994, 1.247), respectively (Fig. 3). A standardized increase of $10 \mu\text{g}/\text{m}^3$ in pre-pregnancy PM_{10} (standardized OR = 1.041, 95% CI: 0.974, 1.108), first trimester (standardized OR = 0.988, 95% CI: 0.972, 1.004), or second trimester (standardized OR = 0.957, 95% CI: 0.895, 1.018) was not associated with GDM risk (Fig. 4). We also found that a $10 \mu\text{g}/\text{m}^3$ increase in pre-pregnancy (standardized OR = 1.012, 95% CI: 0.942, 1.082), first trimester (standardized OR = 1.004, 95% CI: 0.973, 1.035), second trimester (standardized OR = 0.989, 95% CI: 0.950, 1.028) or third trimester NO_2 (standardized OR = 0.984, 95% CI: 0.904, 1.064) was not significantly associated with GDM risk (Fig. 6), respectively.

3.2.4. Publication bias and sensitivity

No substantial publication bias was observed in the analysis of the association between $\text{PM}_{2.5}$, PM_{10} , SO_2 , NO_2 or O_3 exposure and GDM (Supplementary Table 3). In the sensitivity analysis, the pooled estimates for $\text{PM}_{2.5}$, SO_2 , NO_2 or O_3 exposure and GDM were robust (Supplementary Figs. 1–4).

4. Discussion

Evidence from both mechanism and epidemiological studies has provided insights into the association between maternal exposure to air pollution and GDM. Limited mechanism studies indicate that β -cell dysfunction, neurohormonal disturbance, inflammation, oxidative stress, gut microbiome and insulin resistance may be involved in the

relationship between air pollution and GDM. The results of meta-analysis showed that maternal first trimester exposure to SO_2 increased the risk of GDM, while pre-pregnancy exposure to O_3 was inversely associated with GDM risk. No significant effects were observed for $\text{PM}_{2.5}$, PM_{10} and NO_2 .

Although the sources varied between and within countries, the main sources of SO_2 have been known to be transformation industries and energy combustion (Holnicki et al., 2018). The potential biological mechanisms underlying SO_2 exposure and GDM are still unclear. Yang et al. observed that exposure to SO_2 increased the concentrations of fasting glucose, 2 h glucose and 2 h insulin based on a large cross-sectional study in China (Yang et al., 2018b), which provide solid support for the diabetic effects of SO_2 exposure. The increased concentration of fasting glucose may trigger glucose-stimulated insulin secretion to maintain glucose homeostasis (Parsons et al., 1992), which increases the demand for β -cells. It is thought that β -cells deteriorate due to excessive insulin production (Plows et al., 2018), thus exposure to SO_2 may accelerate β -cell dysfunction and then lead to GDM eventually.

O_3 is a strong oxidant that exerts its action by generating reactive oxygen species (Romieu et al., 2010), exposure to O_3 may induce oxidative stress among pregnant women, which is known to mediate the development of insulin resistance (Lappas et al., 2011; Plows et al., 2018). To date, the mechanism of air pollution and the development of GDM remain unknown. Few studies are conducted to explore the responsible biological linkage between air pollution and the development of GDM. Future mechanism studies on the molecular level are needed to provide persuasive rationale underlying the relationship. Why we observed pre-pregnancy exposure to O_3 was inversely associated with GDM risk is not clear. One explanation might be that pregnant women before pregnancy exposed to high levels of O_3 may result in miscarriage before GDM diagnosis. These people were removed from study population so

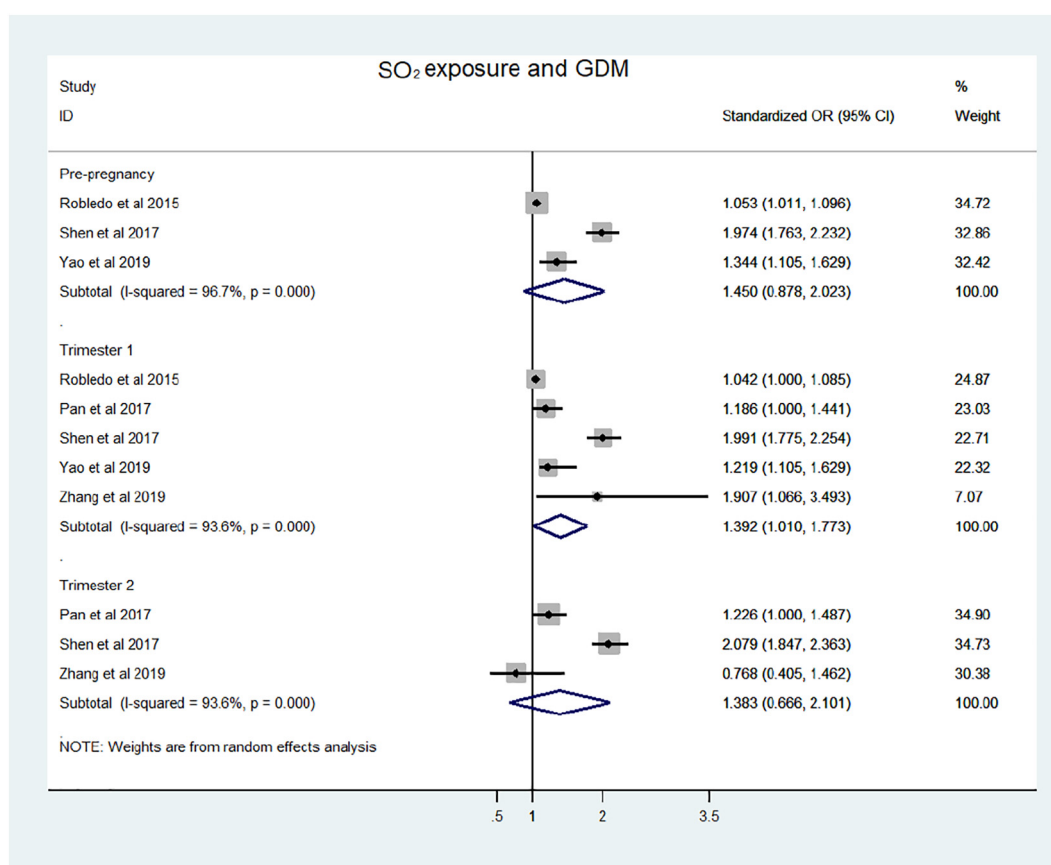


Fig. 5. Standardized odds ratio (OR) and 95% confidence intervals (CI) of GDM associated with 10 µg/m³ increase in SO₂ by gestational period; pooled estimates of effect size are indicated by vertical points of diamonds and 95% CI are represented by horizontal points; size of shaded area around point estimate is proportional to weight in calculating pooled estimate.

that a protective effect of O₃ was observed (Heejoo et al., 2019; Padula et al., 2019).

The first epidemiological study concerning air pollution and GDM was conducted by van den Hooven et al. in the Netherlands in 2009, which observed no effects (van den Hooven et al., 2009). From then on, numerous epidemiological studies have examined the relationship between air pollution exposure and GDM risk. However, results are inconsistent. It is worth noting that inconsistency among these studies may also reflect intrinsic dissimilarities in exposure assessment, outcome definition, effect estimation models and adjusted variables, which may contribute to the heterogeneity.

Air pollution exposure assessment is important for estimating the effect of air pollutants on the risk of GDM. Studies included in the meta-analysis were conducted in different countries. The ambient air pollutant level can be quite different among study locations. There has been some evidence suggesting inconsistent risk estimates for air pollutants due to different air pollution level among locations. Exposure assessment models in selected studies of this meta-analysis were different, some considered spatial as well as temporal variability (Choe et al., 2019; Choe et al., 2018; Fleisch et al., 2014; Fleisch et al., 2016; Hu et al., 2015; Pedersen et al., 2017; Robledo et al., 2015), others didn't (Heejoo et al., 2019; Pan et al., 2017; Shen et al., 2017; Yao et al., 2019; Zhang et al., 2019). What's more, personal exposure assessment methods also differed, which was based on maternal residential address (Choe et al., 2019; Choe et al., 2018; Fleisch et al., 2014; Fleisch et al., 2016; Heejoo et al., 2019; Hu et al., 2015; Malmqvist et al., 2013; Pedersen et al., 2017; Zhang et al., 2019), participants' resident townships (Pan et al., 2017; Shen et al., 2017) or delivery hospital referral region (Robledo et al., 2015), however, measurement error could have occurred for not taking into account the fact that participants have different activity models (time spent at different locations) and residential

mobility during pregnancy. Exposure misclassification among different studies may present from exposure assessment models or personal exposure assessment, which may partly explain the heterogeneity among these studies. Future studies that considering participants' activity models and residential mobility as well as the spatial and temporal variability of air pollution exposure may be warranted to reduce potential exposure misclassification. Furthermore, few studies examined the association between different constituents of particulate matter and the risk of GDM. Only one study found that an IQR increase in first trimester PM_{2.5} was not associated with GDM, but they observed different effects of PM_{2.5} constituents, for example, nitrate was associated with an increased GDM risk while sulfate was associated with a decreased risk of GDM (Robledo et al., 2015).

GDM is routinely screened during 24th to 28th gestational weeks (Association, 2004). Within 13 studies included for this meta-analysis, six studies used American Diabetes Association diagnosis criteria (Fleisch et al., 2014; Fleisch et al., 2016; Heejoo et al., 2019; Hu et al., 2015; Pan et al., 2017; Robledo et al., 2015), three studies used the International Classification of Diseases criteria (Choe et al., 2019; Choe et al., 2018; Shen et al., 2017), one used WHO criteria (Malmqvist et al., 2013). Pedersen et al. used both the International Classification of Diseases criteria and WHO criteria (Pedersen et al., 2017). The WHO criteria has been rarely used in Denmark, so we included the adjusted ORs from the International Classification of Diseases criteria in Denmark, which was very similar to criteria used in Sweden (Malmqvist et al., 2013) where GDM was diagnosed based on plasma glucose >10 mmol/L 2 h after 75 g oral glucose tolerance test during 28th gestational week. For American Diabetes Association criteria, diagnosis of GDM is based on laboratory values confirming a plasma glucose level test at least 2 plasma glucose values meeting or exceeding the following values on the 100-g or 75-g oral glucose tolerance test: fasting, 95 mg/

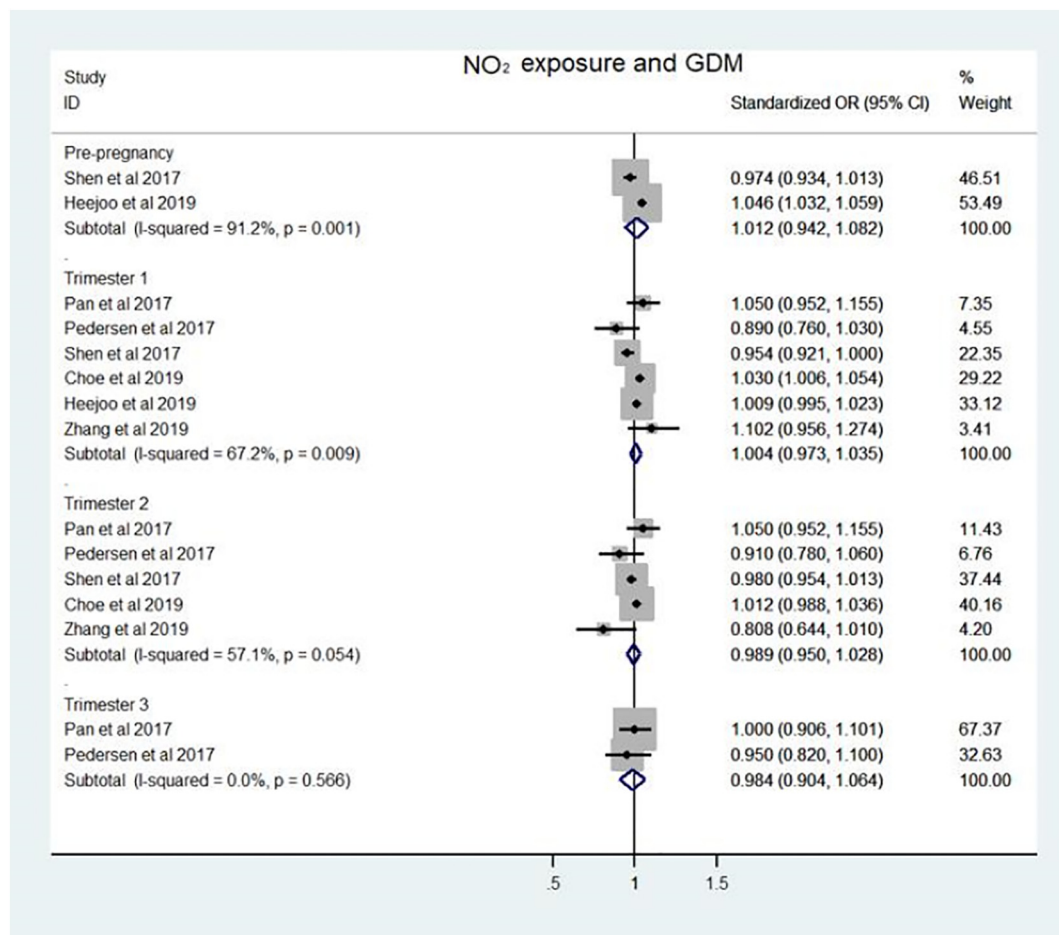


Fig. 6. Standardized odds ratio (OR) and 95% confidence intervals (CI) of GDM associated with 10 µg/m³ increase in NO₂ by gestational period; pooled estimates of effect size are indicated by vertical points of diamonds and 95% CI are represented by horizontal points; size of shaded area around point estimate is proportional to weight in calculating pooled estimate.

dL (5.3 mmol/L); 1 h, 180 mg/dL (10.0 mmol/L); 2 h, 155 mg/dL (8.6 mmol/L); and 3 h, 140 mg/dL (7.8 mmol/L) as previously described (Association, 2004). Studies from America (Choe et al., 2019; Choe et al., 2018) and Taiwan (Pan et al., 2017) diagnosed GDM based on the International Classification of Diseases, which usually used American Diabetes Association diagnosis criteria to diagnose GDM. Yao et al. used the Chinese GDM criteria (fasting, 5.1 mmol/L; 1 h, 10.0 mmol/L; 2 h, 8.5 mmol/L) (Yao et al., 2019), which is similar to Zhang et al.'s study (Zhang et al., 2019) using the IADPSG criteria (for a 75-g oral glucose tolerance test at 24–28 week of gestation, normal values: 0 h < 5.1 mmol/L; 1 h < 10 mmol/L; 2 h < 8.5 mmol/L, if one or more values exceeded the threshold, then it will be diagnosed with GDM) (Lapolla et al., 2011). With any approach, the diagnosis of GDM is based on an oral glucose tolerance test during 24th–28th week of gestation.

Generally, the third trimester of pregnancy is from the 28th week of gestation to delivery (Wang et al., 2018b). However, the oral glucose tolerance test is routinely screened during 24th to 28th weeks of gestation, it might be inappropriate to average the exposure including the third trimester. The ranges of trimesters differed among studies, for example, second trimester in some studies ranged from 13th to 24th weeks of gestation (Fleisch et al., 2016; Shen et al., 2017), 13th to 26th weeks of gestation (Choe et al., 2019; Choe et al., 2018; Pan et al., 2017) or from 14th to 26th weeks of gestation (Hu et al., 2015). As mentioned above, GDM is routinely screened between 24th and 28th weeks of gestation. The information on when GDM is diagnosed was not collected in current studies. It may be inappropriate to include the exposure after GDM diagnosis.

Physiological changes throughout gestation generally occur week by week, including endocrinology, the cardiovascular system, the respiratory system and water balance (Descamps et al., 2000; Kohlhepp et al., 2018; Zdolska-Wawrzkiwicz et al., 2019). Those changes are necessary for the pregnant women to adapt to this physiologic event. It is hypothesized that the sensitivity of ambient air pollution exposure for pregnant women may vary throughout pregnancy (Xia et al., 2019). Most studies fitted the regression model for each trimester when assessing the relationship between air pollution exposure and GDM risk (Choe et al., 2018; Fleisch et al., 2014; Fleisch et al., 2016; Hu et al., 2015). It may be insufficient to identify susceptible window trimester-specifically due to the biological response to air pollution may not align with trimesters exactly. Wilson et al. demonstrated that effect estimates at trimester level may be biased and identified incorrect windows (Wilson et al., 2017). Considering air pollution is time-varying and may cause delayed effects, time-series oriented analytical methods such as distributed lag models exploring the relationship between air pollution exposure and GDM risk may be warranted.

Recently, GDM is more recognized by the scientific community, but the mechanisms involved in GDM are complex and not been fully understood (Chiefari et al., 2017). Lifestyle behaviors including diet, physical activity, smoking, weight gain, maternal age, BMI and stress during pregnancy have been associated with GDM (Alves et al., 2019; Goldstein et al., 2017; Lao et al., 2006; Schwartz et al., 2015; Zhang et al., 2014). These factors may be the driving forces behind the increasing prevalence of GDM observed in some countries (Jeppesen et al., 2017). The relationship between air pollution and GDM might be confounded and

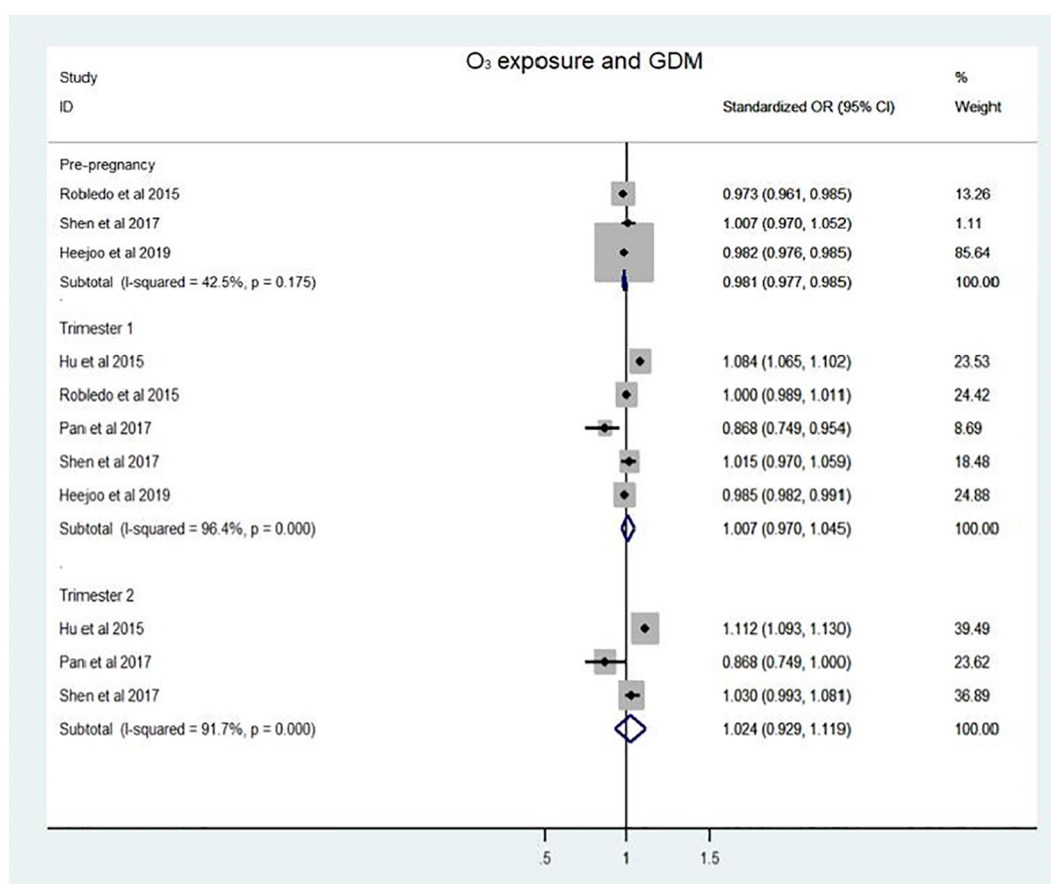


Fig. 7. Standardized odds ratio (OR) and 95% confidence intervals (CI) of GDM associated with $10 \mu\text{g}/\text{m}^3$ increase in O_3 by gestational period; pooled estimates of effect size are indicated by vertical points of diamonds and 95% CI are represented by horizontal points; size of shaded area around point estimate is proportional to weight in calculating pooled estimate.

modified by these potential factors, which should be taken into account in the future studies. However, data collected in most studies were from the Department of Health, hospitals or from Birth Registry System, so that some important factors such as maternal lifestyle behaviors were not adjusted when assessing the air pollution-GDM relationship. More prospective cohort studies are needed, especially in developing countries to support robust recommendation. What's more, future studies that assess the single and joint effects of different air pollutants, other related exposures including road traffic noise, green space and meteorological factors, such as temperature and humidity are warranted.

Several limitations in our study should be noted. First, the number of the epidemiological studies included in the meta-analysis is limited, especially for NO_2 in the first trimester and PM_{10} in the second trimester. However, we evaluated the quality of included studies using the Newcastle-Ottawa Scale (NOS) (Li et al., 2017; Wells et al., 2013) and found all epidemiological studies included were of high quality (score ranged from 7 to 8 in NOS). Moreover, standardized increments for all pollutants with different concentrations in included studies were used (Liu et al., 2019a; Yang et al., 2018a). Further, we have examined the publication bias and no substantial publication bias was observed, therefore, results of this meta-analysis are convincing. Second, high heterogeneity exists, therefore the random effect models were used to pool the estimates of studies with substantial heterogeneity in our meta-analysis, however, limitations of this random model in underestimating the statistical error and yielding overconfident conclusions exist (Doi et al., 2015), so that the precision of our meta-analysis might be compromised. Third, most published papers were derived from developed countries like the United States and other European countries. The number of studies from developing countries with high air pollution concentrations like India is limited. Finally, associations between other

pollutants (such as black carbon, CO or NO_x) and GDM were not assessed in our study due to limited relevant studies (<2).

5. Conclusions

Growing existing evidence from both mechanism and epidemiological studies provided insights into the association between maternal exposure to air pollution and GDM. Limited mechanism studies indicated that β -cell dysfunction, neurohormonal disturbance, inflammation, oxidative stress, gut microbiome and insulin resistance may be involved in the relationship between air pollution and GDM. However, no mechanism studies were conducted to explore the direct biological linkage between air pollution and GDM. Additional studies on the molecular level are needed to provide persuasive rationale underlying the relationship. The results of meta-analysis showed that maternal first trimester exposure to SO_2 increased the risk of GDM, while pre-pregnancy exposure to O_3 was inversely associated with GDM risk. No significant effects were observed for $\text{PM}_{2.5}$, PM_{10} and NO_2 . Maternal lifestyle, and other risk factors of GDM that may modify the air pollution-GDM relationship should also be considered in future studies. More prospective cohort studies are warranted in developing countries with high levels of air pollution.

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Authors' contributions

H.Z. and C.H. conceived and designed the study, H.Z. and Q.W. collected and extracted the data, H.Z. performed the statistical analysis, H.Z. drafted the initial version of the manuscript, C.H., Y.Z. and D.J. reviewed and helped to write the second version of the manuscript. S.H., K.W., M.R. and H.D. helped to revise the manuscript. All authors read and approved the final manuscript.

Declaration of competing interest

No potential conflicts of interest relevant to this article were reported.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2020.137349>.

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